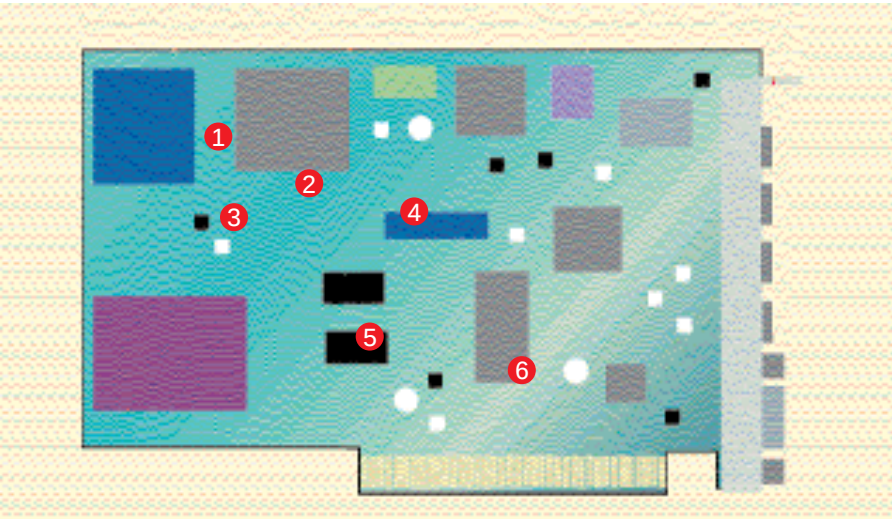




Buying Tips

- ✓ **DEFINE YOUR USAGE:** If you love watching DVDs, and don't play games too often, opt for a sound card that supports multiple channels, such as 5.1 surround. If you want a gaming solution, buy a card that supports the standard 3D sound APIs, such as DirectSound, EAX, A3D and I3DL2.
- ✓ **WARRANTY:** Make sure you get at least a one-year warranty for budget sound cards—under Rs 2,000—and at least a three-year warranty for more expensive ones.
- ✓ **INTERFACE:** Make sure that the sound card's interface is PCI and not ISA.
- ✓ **DRIVERS:** Check that the sound card you're buying has the drivers bundled, especially if you are a non-Windows user.
- ✓ Audiophiles should look for gold-plated connectors for minimum signal loss.
- ✓ Check for bundled applications such as games and media players.
- ✓ Make sure that the sound card has at least two available speaker connectors for connecting surround sound speakers. If not, make sure that there is an alternative provision for connecting 5.1 or 6.1 speakers.
- ✓ Make sure that the sound card is full-duplex, which means that it can record and play sound at the same time.
- ✓ Buy a sound card that has low processor usage, otherwise listening to music and working on a resource intensive application simultaneously will not be a pleasant experience.

- 1 **DSP:** The Digital Signal Processor processes, converts and analyses signals, and does other computational work as well.
- 2 **DAC:** The DAC is responsible for all audio signals leaving the computer.
- 3 **ADC:** The ADC is responsible for all audio signals coming into the computer.
- 4 **Connectors:** The MIDI connector is used to connect to an external music instrument, such as an electronic keyboard. A game port is used to connect joysticks and other gaming devices, while other connectors include the line-in, line-out and microphone jacks. Each of these is colour-coded. On-board connectors include the CD-audio connector and the S/PDIF connector.
- 5 **Wave table synthesizer:** This contains

Quick Find	
YOU NEED	A basic, no-frills sound card
Basic Use	
LOOK FOR	A full-duplex card, or a motherboard with an on-board sound chip that provides simple stereo or 2.1 channel output
YOU NEED	Surround sound at an affordable price
Inter Use	
LOOK FOR	A card with 4.1 or 5.1 channel support; in-built Dolby Digital decoding will be an added bonus
YOU NEED	Excellent sound quality for professional level editing
Prof Use	
LOOK FOR	A card that supports 24-bit recording at 96 KHz, along with S/PDIF, DIN, or FireWire connectors

tiny sample recordings of actual instruments. These sounds can be played back through the sound card accurately.

6 **Amplifier:** The amplifier amplifies the audio signals, and passes it on to the line-out of the sound card.

Do Remember

- Some sound cards only have FM synthesizers, and these are considered a bad choice for creating quality music. Professionals should buy a sound card that features Wavetable and SoundFont technology support.
- S/PDIF—Sony/Philips Digital Interface—is a standard for transmitting data in a lossless digital format, to preserve sound quality. Sound cards that feature an S/PDIF connector are therefore a better choice.
- When listening to a set of stereo speakers or headphones, the left ear may hear sounds meant for the right ear, and vice versa. A well engineered sound card—one with gold-plated connectors and shielded wires, for instance—will suppress this interference, and produce more accurate sound.

Facts

- Integrated sound chips use a lot of system resources. They make good makeshift solutions, but generally offer lower quality output as compared to standalone cards.
- Check the specifications of the sound cards you are interested in against the best available. The SNR ratio (dB), THD (%) and the amount of cross-talk are three major factors to look for. This will help you make a decision.



The Future

Sound cards don't see changes very frequently, but when they do, it's generally something significant. Labs are already working on 9.1-channel surround sound. This increase in the number of channels, and hence, speakers, will not only make future movie experiences more immersive, but also give hardcore gamers an edge over rivals by allowing them to predict more precisely where an attack is coming from.